DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS
First Semester, SY 2024-2025

COURSE CODE	:	S-ITPC315
COURSE TITLE	:	Networking 2
COURSE TYPE	:	Lecture
COURSE CREDIT	:	Two Units
LEARNING MODALITY	:	Traditional Blended Learning
PRE-REQUISITES	:	S-ITCS214 and S-ITCS214/LA
CO-REQUISITES	:	S-ITPC315LA
PROFESSOR	:	Joneil M. Baculod Orlando A. Zapanta Cesar Galingana Alrnold Galve
CONSULTATION HOURS	:	Saturday: 7AM – 8AM MS Teams Thursday: 9AM – 12NN MS Teams
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VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

Deals with both static routing and dynamic routing protocols. By examining multiple routing protocols, you will gain a better understanding of each of the individual routing protocols and a better perspective of routing in general. Learning the configuration of routing protocols is fairly simple. Developing an understanding of the routing concepts themselves is more difficult, yet is critical for implementing, verifying, and troubleshooting routing operations

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO 1. Create a fully connected LAN

CLO 2. Explain the principles of IPv4 and IPv6 address management.

CLO 3. Configure basic network security.

CLO 4. Explain how to use security best practices to mitigate attacks.

CLO 5. Analyse routing algorithms protocols, process routing tables and configure routers

for proper orientation of an efficient network.

CLO 6. Create and implement static routes IPv4 and IPv6.

CLO 7. Create and implement dynamic routing protocols.

CLO 8. Test for common configuration issues.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
Midterm Period	
CLO1. Create a fully connected LAN.	TLO1. Explain the need for routing.
CLO2. Explain the principles of IPv4 and IPv6 address management.	TLO2. Explain the principles of IPv4 and IPv6 address management.
CLO3. Configure basic network security.	TLO3. Configure devices for secure remote management.
CLO4. Explain how to use security best practices to mitigate attacks.	TLO4. Explain how security tools and software updates mitigate network security threats. TLO5. Explain how antimalware software mitigates data loss and service disruptions.

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours

Module 1 CLO 1 CLO 3 TLO 1 TLO 6	Week 1 Aug 19-24	• Gospel reading reflection and prayer • Presentation on the synchronous flow and activities to do during asynchronous mode. • LMS Expectations: • Netacad Course Experience – Interface, Submission, Around the Online Module • Schoolbook Details for Submission of Assessment • Module 1: Routing Between Networks • Objective: Create a fully connected LAN • Define the new key terms in this module • Discussion with following 2-3 mins. videos ○ Dividing the Local Network ○ Router Packet Forwarding ○ Messages Within and Between Networks Part 1 ○ Messages Within and Between Networks Part 2 ○ Messages Sent to Remote Networks • Formative Activity: Check Your Understanding ○ IP Packet Encapsulated in an Ethernet Frame ○ Path Selection ○ Packet Forwarding ○ Select the Default Gateway ○ The Routing Table • Share the screen cap of your achievement in the quiz • Download Packet Tracer version 8.0.1	MS Teams NetAcad	✓		2
Module 2 CLO 2 TLO 2	Week 2 Aug 26-31	• Gospel reading reflection and prayer • Recap of previous meeting achievements • Module 2: The Internet Protocol • Objective: Explain the features of an IP address ○ Define the new key terms in this module ○ Discussion on IPv6 replacing IPv4 ○ Subnetting Networks – Classful and Classless IPv4 ○ Explain Features of IPv6 • Formative Activity: Check Your Understanding ○ Calculate Whether the Destination is Local or Remote ○ IPv4 Address Structure ○ Classful IPv4 Address • Enabling Assessment 1: Score in the PT Activity	MS Teams NetAcad		✓	2

Module 2 CLO 2 TLO 2	Week 3 Sept 2-7	• Gospel reading reflection and prayer • Module 2 continuation • Module 2: The Internet Protocol • Objective: Explain the features of an IP address o Define the new key terms in this module o Discussion on IPv6 replacing IPv4 o Student Presentation of Subnetting Skills • Enabling Assessment 2: Score in the PT Activity	MS Teams NetAcad			2
Module 3 CLO 1 CLO 4 TLO 4 TLO 5 TLO 6	Week 4 Sept 9-14	• Gospel reading reflection and prayer • Read Module 3: Cisco Switches and Routers • Objective: Compare in-band and out-of-band management access • Describe LAN switches, switch boot process, small business routes and the router boot process. • You may try the module provided videos and animations • Enabling Activity 3 – Module3 Handwritten New Terms and Concepts	MS Teams NetAcad	✓		2
Module 3 CLO 1 CLO 4 TLO 4 TLO 5 TLO 6	Week 5 Sept 16-21	• Gospel reading reflection and prayer • Read Module 3: Cisco Switches and Routers continuation • Objective: Compare in-band and out-of-band management access • Describe LAN switches, switch boot process, small business routes and the router boot process. • You may try the module provided videos and animations Enabling Activity 3 – Module3 Handwritten New Terms and Concepts			✓	2
Module 4 CLO 4 TLO 4 TLO 5	Week 6 Sept 23-28	• Gospel reading reflection and prayer • Read Module 4: The Cisco IOS Command Line • Objective: Explain how to use the Cisco IOS. • Summative Assessment 1 is now available online	MS Teams NetAcad	✓		2

Module 4 CLO 1 TLO 1 TLO 2 TLO 3	Week 7 Sept 30 - Oct 5	• Gospel reading Gospel reading reflection and prayer • Module 4 continuation • Read Module 4: The Cisco IOS Command Line • Objective: Explain how to use the Cisco IOS. • You may try the module provided videos and animations • Formative Activity: ○ Packet Tracer – Navigate the IOS ○ Packet Tracer – Use Cisco IOS Show Commands • Enabling Activity 4 – Module4 Handwritten New Terms and Concepts	MS Teams NetAcad		✓	2
Module 5 CLO 1 CLO 2 TLO 1 TLO 2 TLO 3 TLO 4	Week 8 Oct 7-12	• Gospel reading reflection and prayer • Read Module 5: Build a Small Network • Objective: Build a simple computer network using Cisco devices. • Formative Activity: ○ Packet Tracer – Implement Basic Connectivity ○ Packet Tracer - Configure Initial Router Settings • Enabling Activity 5 – Module5 Handwritten New Terms and Concepts	MS Teams NetAcad	✓		2
Modules 1 -5	Week 9 Oct 14-19	Summative Assessment Midterm Week Exam		✓		2
TOTAL				10	8	18

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
Final Period	
CLO5. Analyze routing algorithms protocols, process routing tables and configure routers for proper orientation of an efficient network.	TLO6. Explain how a router learns about remote networks when operating in a small to medium-sized business network
CLO6. Create and implement static routes IPv4 and IPv6.	TLO7. Configure static routes to enable connectivity in a small to medium-sized business network.

CLO7. Create and implement dynamic routing protocols.	TLO8. Explain the function of dynamic routing protocols.
CLO8. Test for common static and default route configuration issues.	TLO9. Describe some of the approaches used to troubleshoot networks . TLO10. Test for static and default route configurations.

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**	**Onsite / F2F**	**Asynchronous**	**Allotted Hours**
Module 6, 7 CLO 5 CLO 6 TLO 6 TLO 7	Week 10 Oct 21-26	• Gospel reading reflection and prayer • Module 6, 7 Routing Concepts, Static Routing • Objective: Explain how routers use information in packets to make forwarding decisions • Check Your Understanding on Path Determination, Packet Forwarding, IP Routing Table and Compare Static Routing vs. Dynamic Routing • Understand the concept of Static, Default and Floating static route for IPv4 and IPv6 o Enabling Activity 7 – Teacher-made PT on Static Routing	MS Teams NetAcad		✓	2
	Week 11 Oct 28 -Nov 2	***Independent Learning Week***			✓	
Module 8 CLO 7 TLO 6 TLO 8	Week 12 Nov 4-9	• Gospel reading reflection and prayer • Module 8 Dynamic Routing EIGRP • Objective: • Explain EIGRP Characteristics • Implement EIGRP for IPv4 • Explain how EIGRP operates in small to medium-sized networks • Implement EIGRP for IPv6	MS Teams NetAcad	✓		2

Module 9 CLO 7 TLO 6 TLO 8	Week 13 Nov 11-16	• Gospel reading reflection and prayer • Module 9 Dynamic Routing OSPF • Objective: • Explain how single-area OSPF Operates • Implement single-area OSPFv2 • Implement single-area OSPFv2 • Explain how multiarea operates • Implement multiarea OSPF • Formative Activity: o Configure OSPFv2 in a Single Area o Configure OSPFv3 in a Single Area o Configure multiarea OSPFv2 o Configure multiarea OSPFv3 • Enabling Activity 9: Teacher-made PT on OSPF	MS Teams NetAcad		✓	2
Module 9 CLO 7 TLO 6 TLO 8	Week 14 Nov18-23	• Gospel reading reflection and prayer • Module 9 Dynamic Routing OSPF continuation • Objective: • Explain how single-area OSPF Operates • Implement single-area OSPFv2 • Implement single-area OSPFv2 • Explain how multiarea operates • Implement multiarea OSPF • Formative Activity: o Configure OSPFv2 in a Single Area o Configure OSPFv3 in a Single Area o Configure multiarea OSPFv2 o Configure multiarea OSPFv3 Enabling Activity 9: Teacher-made PT on OSPF	MS Teams NetAcad	✓		2
Module 10 CLO 6 TLO 9 TLO 10	Week 15 Nov 25-30	• Gospel reading reflection and prayer • Module 10 Troubleshoot Static and Default Routes	MS Teams NetAcad		✓	2

		• Objective: Troubleshoot static and default route configurations. • Enabling Activity 10: Module6,7 Handwritten New Terms and Commands • Summative Assessment 1 is now available online				
Module 11 CLO 6 CLO 7 TLO 6 TLO 7 TLO 9 TLO 10	Week 16 Dec 2-7	• Gospel reading reflection and prayer • Module 11 Troubleshoot Common Network Problems • Objective: Troubleshoot basic network connectivity issues Enabling Activity 11: Module 8 Handwritten New Terms and Commands	MS Teams NetAcad	✓		2
Module 12 CLO 3 CLO 7 CL 8 TLO 3 TLO 9 TLO 10	Week 17 Dec 9-14	• Gospel reading reflection and prayer • Module 11 Security Considerations • Objective: Explain how to use security best practices to mitigate attacks • Enabling Activity 12: Module 9 Handwritten New Terms and Commands	MS Teams NetAcad		✓	2
Modules 6-12	Week 18 Dec 16-18	Summative Assessment Finals Week Exam		✓		2
TOTAL				10	8	18

GRADING SYSTEM:

Midterm	**%**	**Final**	**%**
Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.
2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.
3. Communication and Feedback

a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.

b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
    a. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
    b. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
    c. Email – you may email your professor. (Professor's email will be given during the course orientation)

c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.
5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).
6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

On-Site References

Provider	Reference Material
QA 76.73.P98 .Or82 2019	Sarker, Dr. M. 2019. Learning Python networking : a complete guide to build and deploy strong networking capabilities using Python 3.7 and Ansible. Second edition. UK : Packt Publishing, Ltd.
IEEE Xplore	Chadha, D. 2019 Software-Defined Optical Networks, IEEE. Wiley-IEEE Press
IEEE Xplore	Panek,C. 2020. Networking Fundamentals. Wiley. Wiley Data and Cybersecurity.

Online References

Call Number/ E-provider	Reference Material
www.netacad.com	(N.d.) Cisco Networking Academy[Accessed : 19 August 2024] Retrieved fromhttps://www.netacad.com/portal/resources/course-resources/ccna-itn

Prepared: Information Technology Department
1st Semester 2024-2025

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